CentraleSupélec

Network Science Analytics

Option Applied Math - M.Sc. in DSBA – M.Sc. in AI

Lecture 1A

Introduction to network science and graph mining

Fragkiskos Malliaros

(Lecture given by George Panagopoulos, PhD student)

Friday, February 7, 2020

Office Hours



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Or, send me an email and we will find a good time to meet

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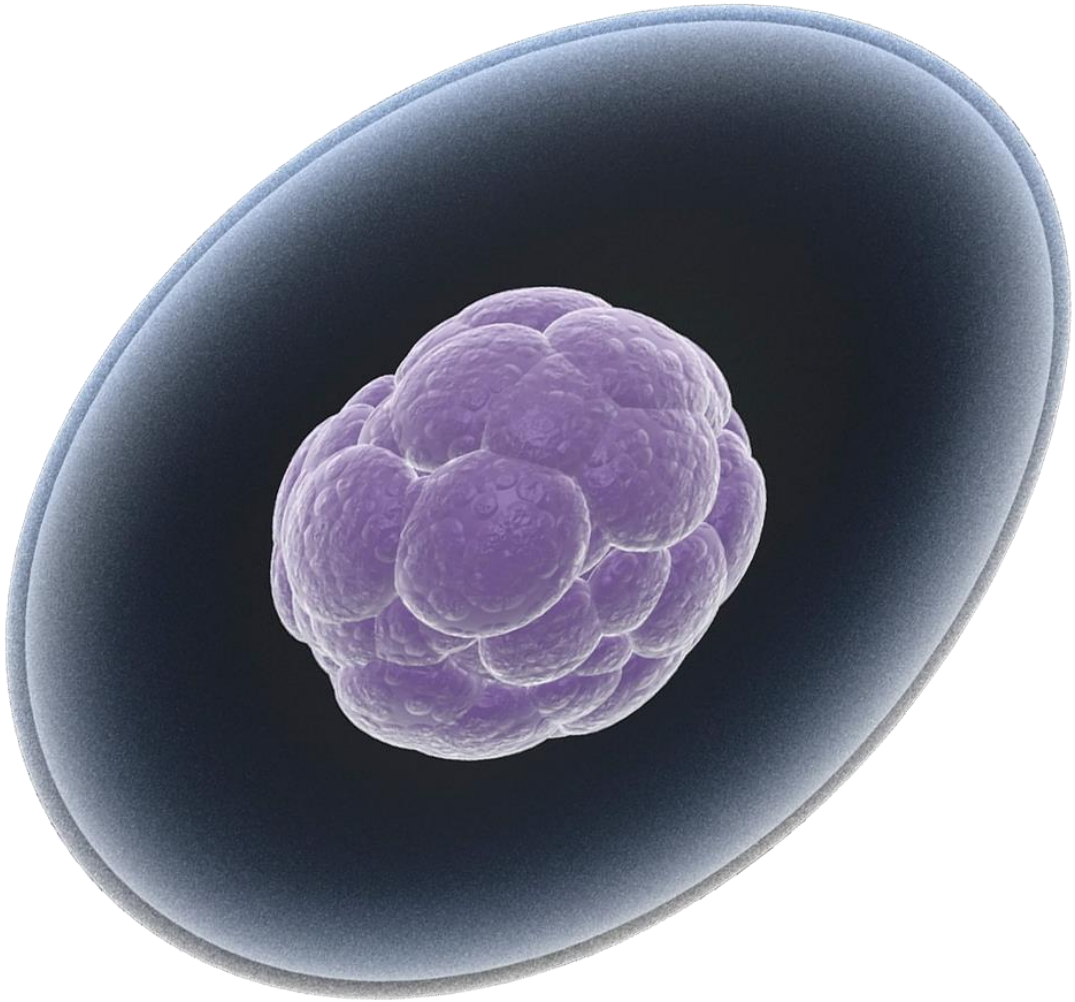
Acknowledgements

- Part of the lecture is based on material by
  - Jure Leskovec, Stanford University
  - Manos Papagelis, York University

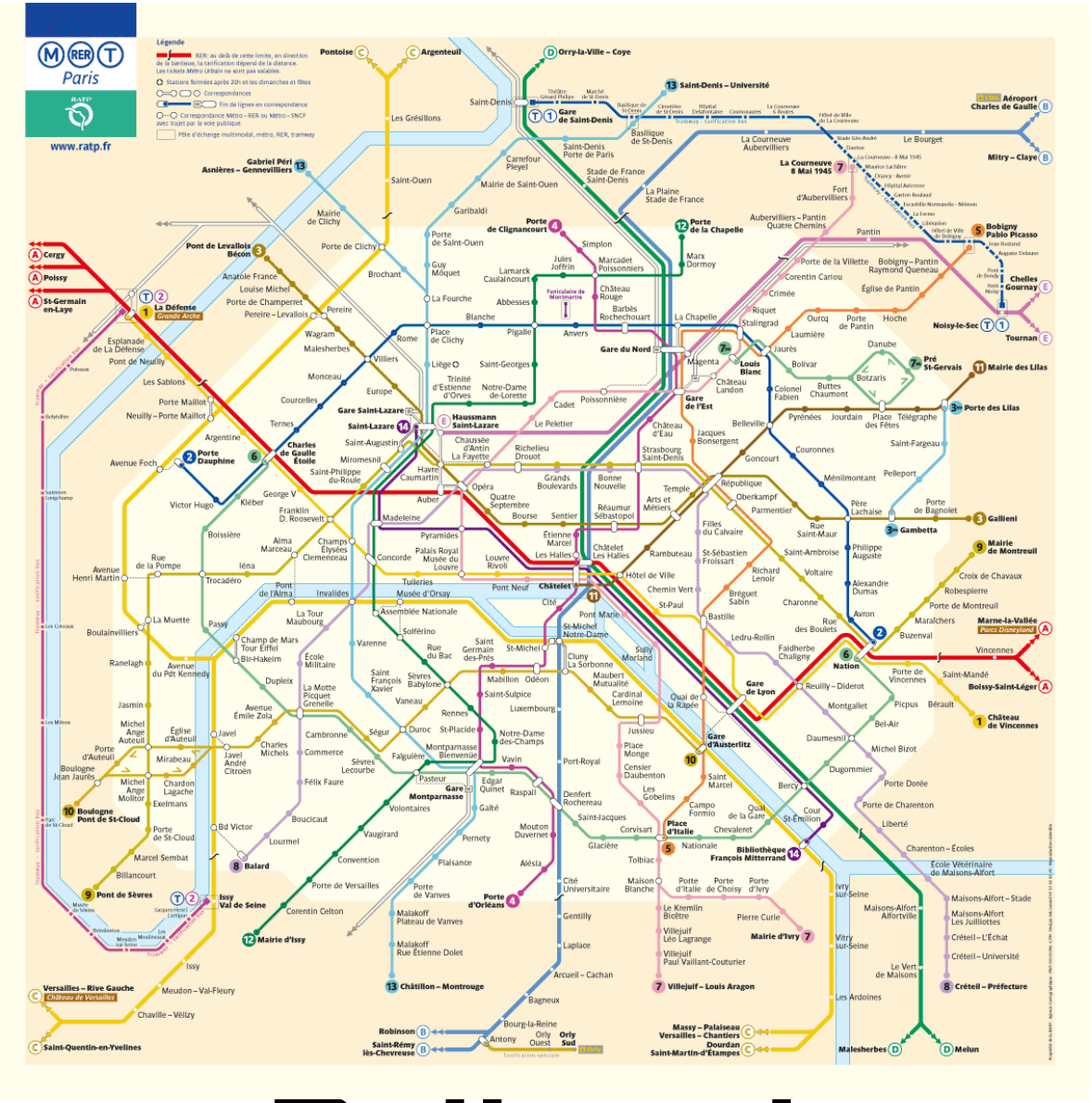
  Thank you!

What do the following things have in common?


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NASDAQ
NASDAQ



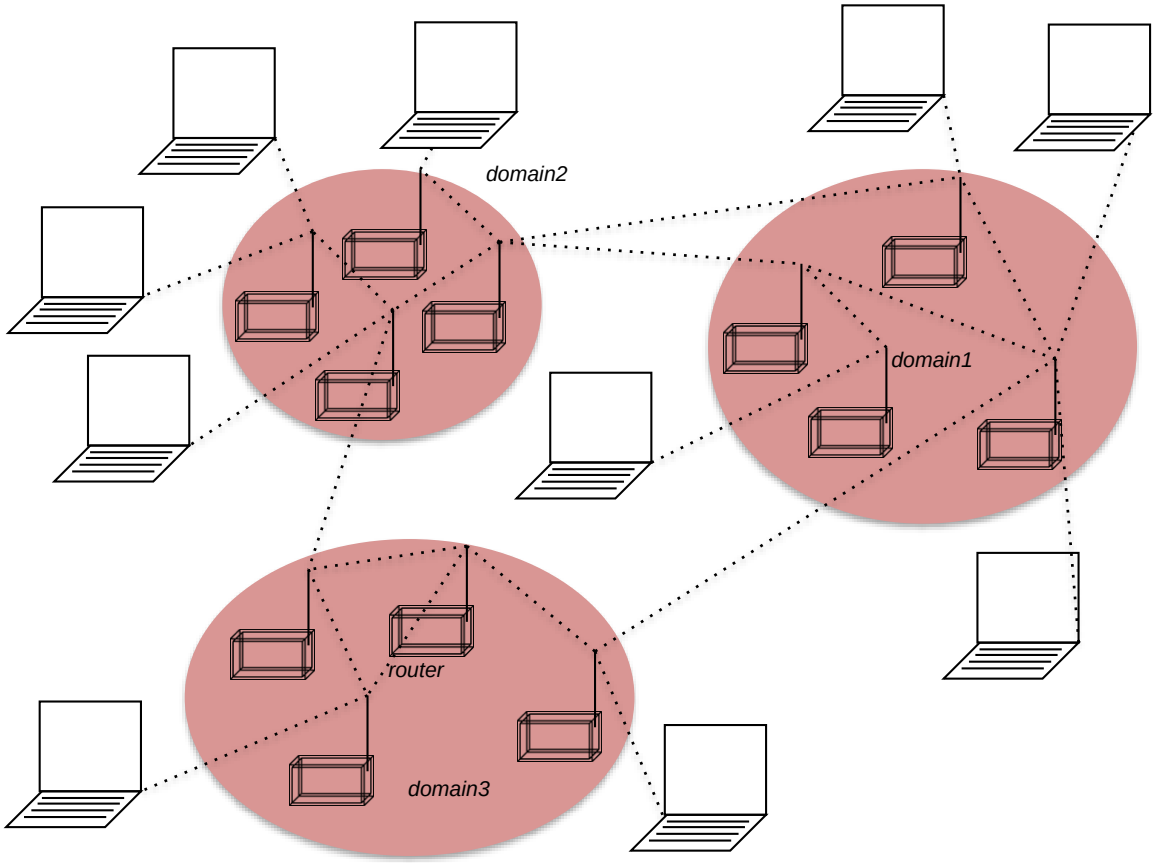
Human cell


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Saint-Denis – Université
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Mitry – Claye
Gabriel Péri Asnières – Gennevilliers
Les Grésillons
Saint-Ouen
Saint-Denis
Gare de Saint-Denis
Basilique de St-Denis
Carrefour Pleyel
Mairie de Saint-Ouen
Stade de France Saint-Denis
La Plaine Stade de France
La Courneuve Aubervilliers
La Courneuve 8 Mai 1945
Le Bourget
Fort d'Aubervilliers
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Pantin
Bobigny Pablo Picasso
Chelles Gournay
Noisy-le-Sec
Tournan
Porte de la Villette
Bobigny – Pantin Raymond Queneau
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Corentin Cariou
Crimée
Riquet
Stalingrad
Ourcq
Laumière
Porte de Pantin
Hoche
Danube
Botzaris
Buttes Chaumont
Colonel Fabien
Bolivar
Jaurès
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Mairie des Lilas
Porte des Lilas
Pyrénées
Jourdain
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Cergy
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Esplanade de La Défense
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Railroads



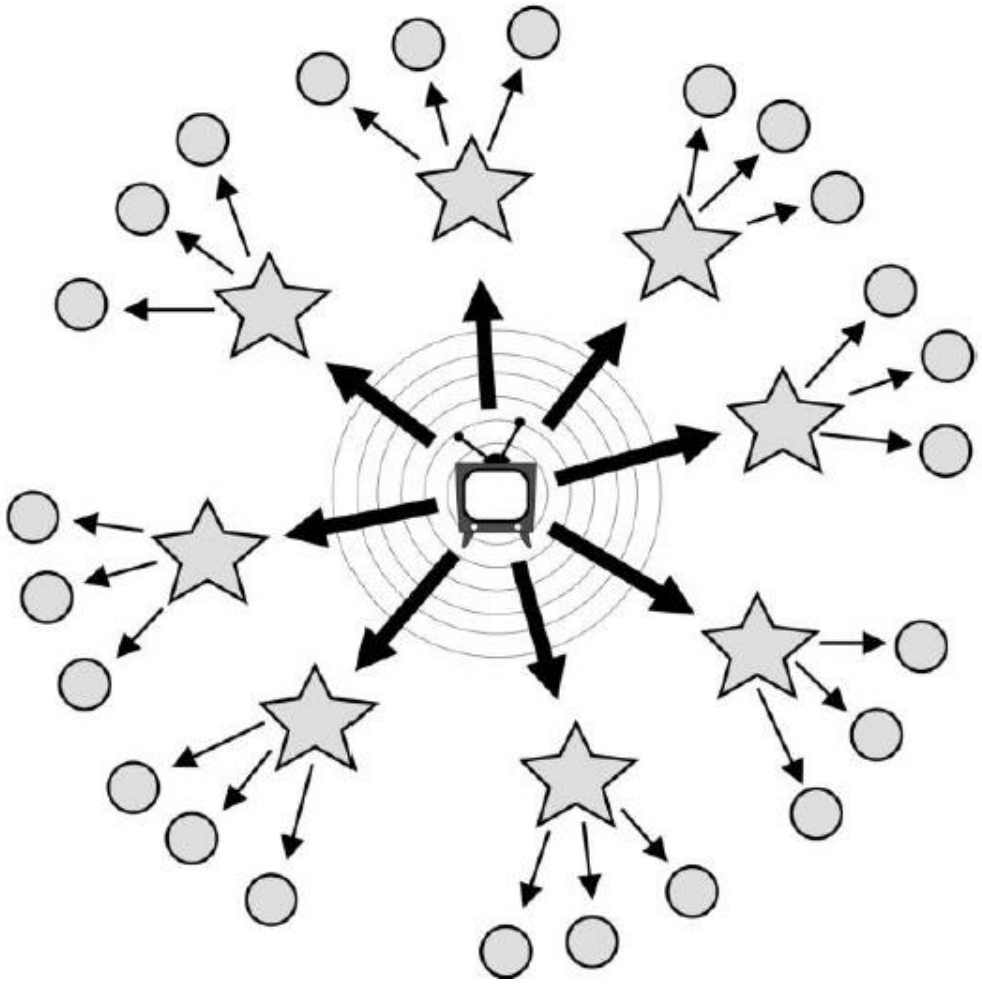
Brain



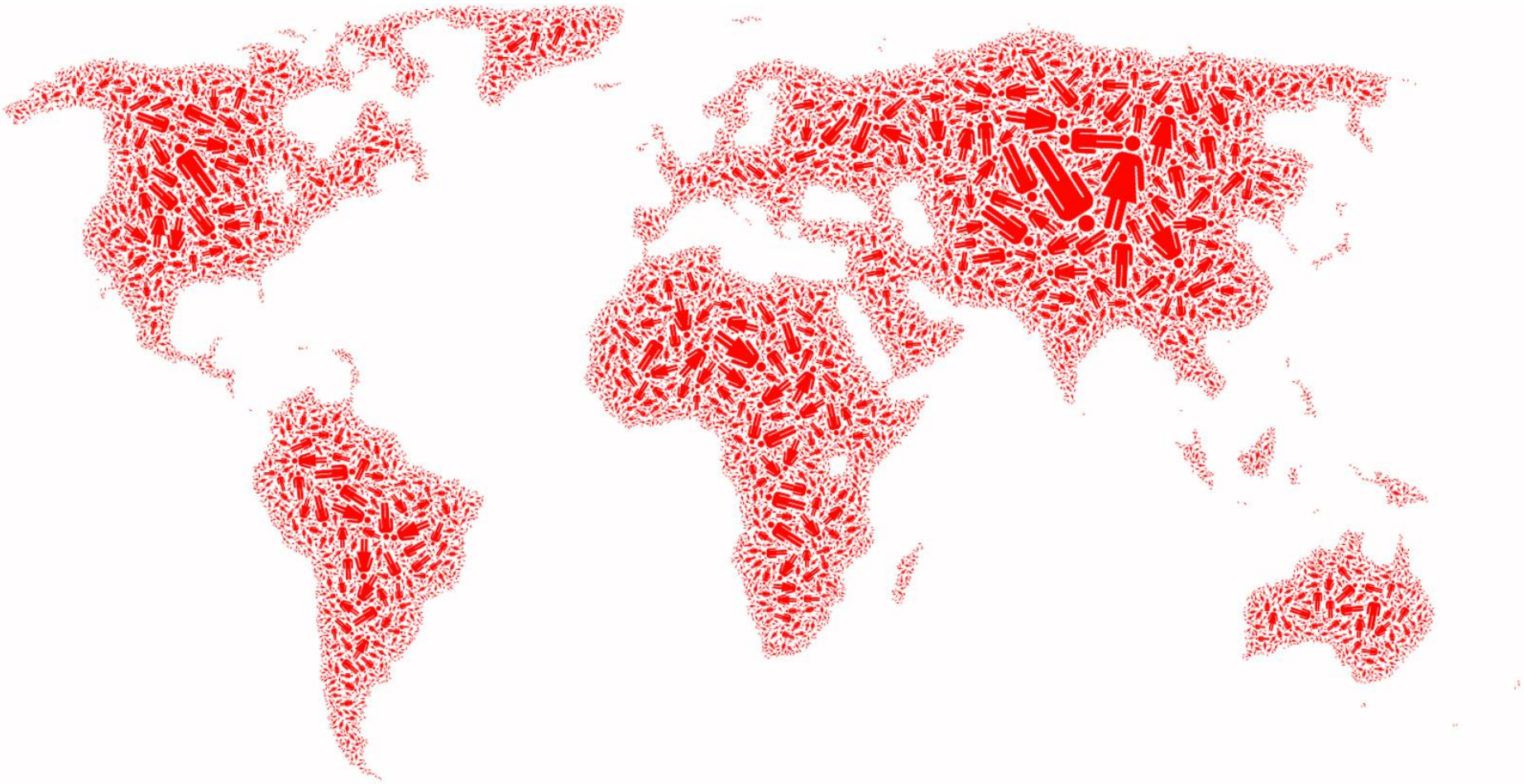
Internet



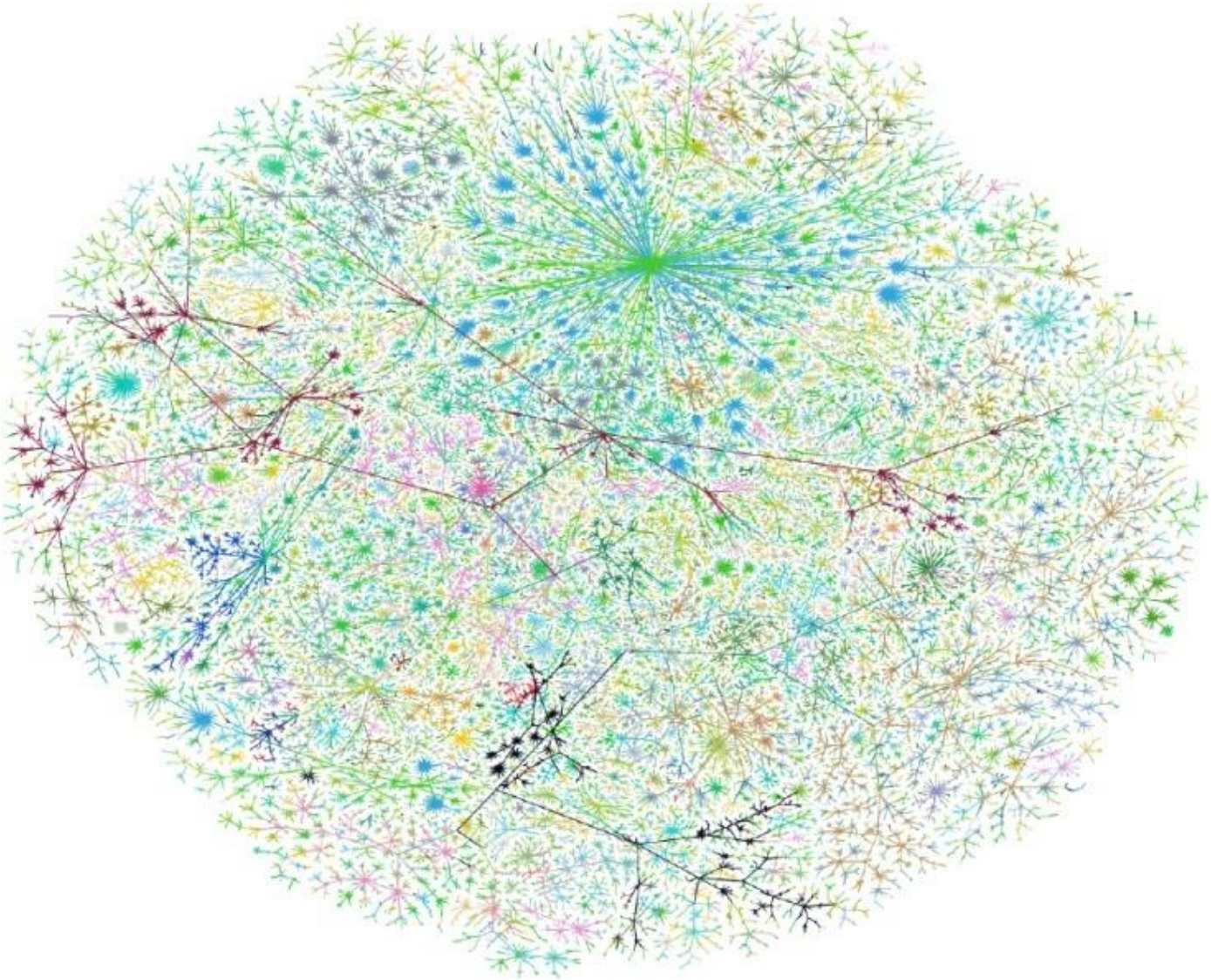
Friends & Family



Media & Information



Society

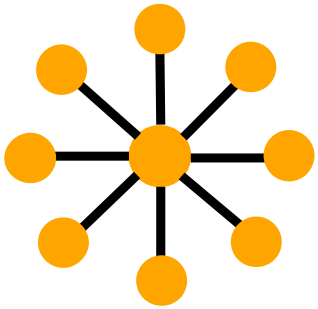


The Network!

Networks



Networks allow to model relationships between entities



General-purpose language

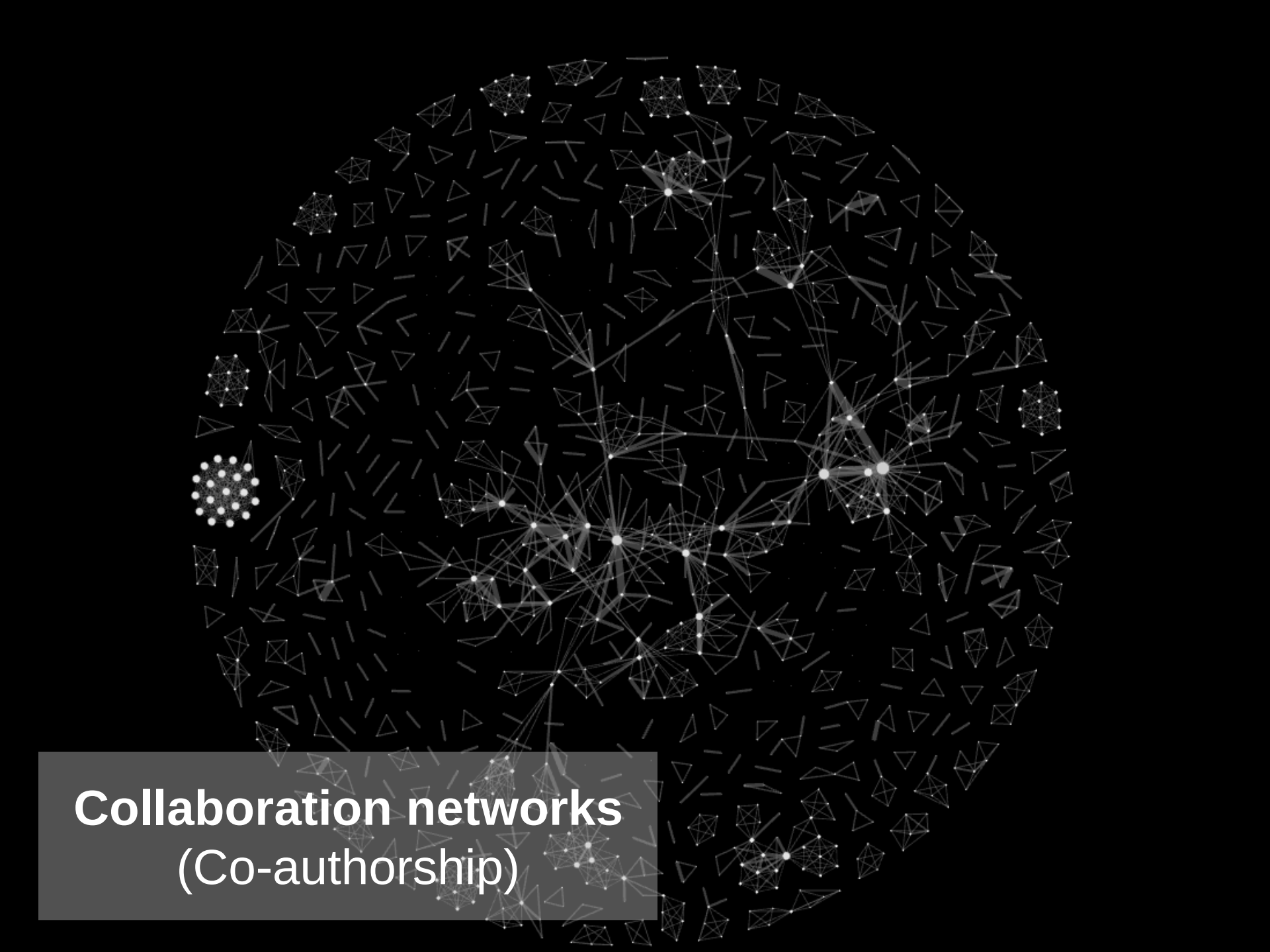
for describing real-world systems



Facebook

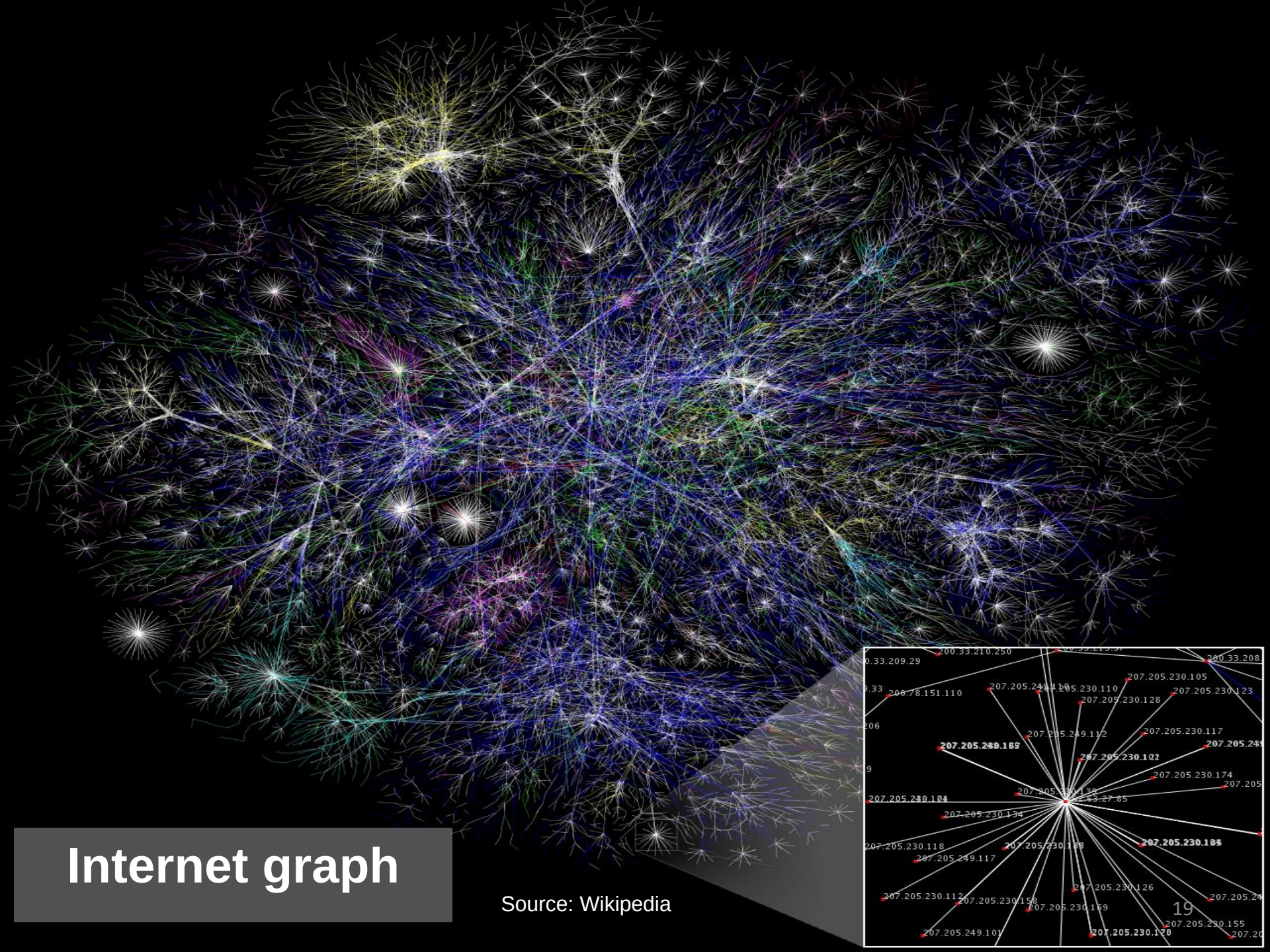
2.07 Billion users (Q2 of 2017)

Source: https://www.facebook.com/zuck

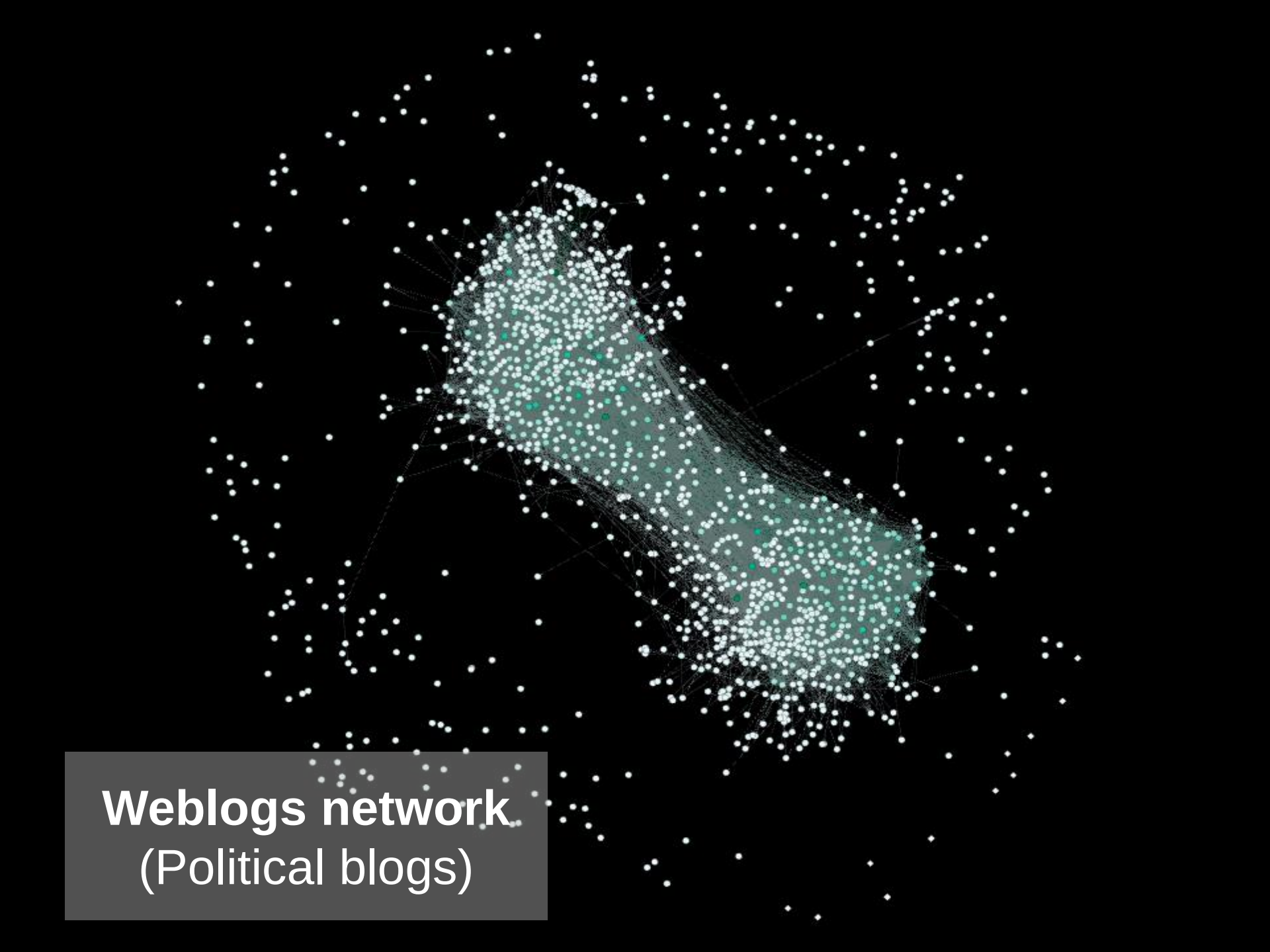


Collaboration networks
(Co-authorship)

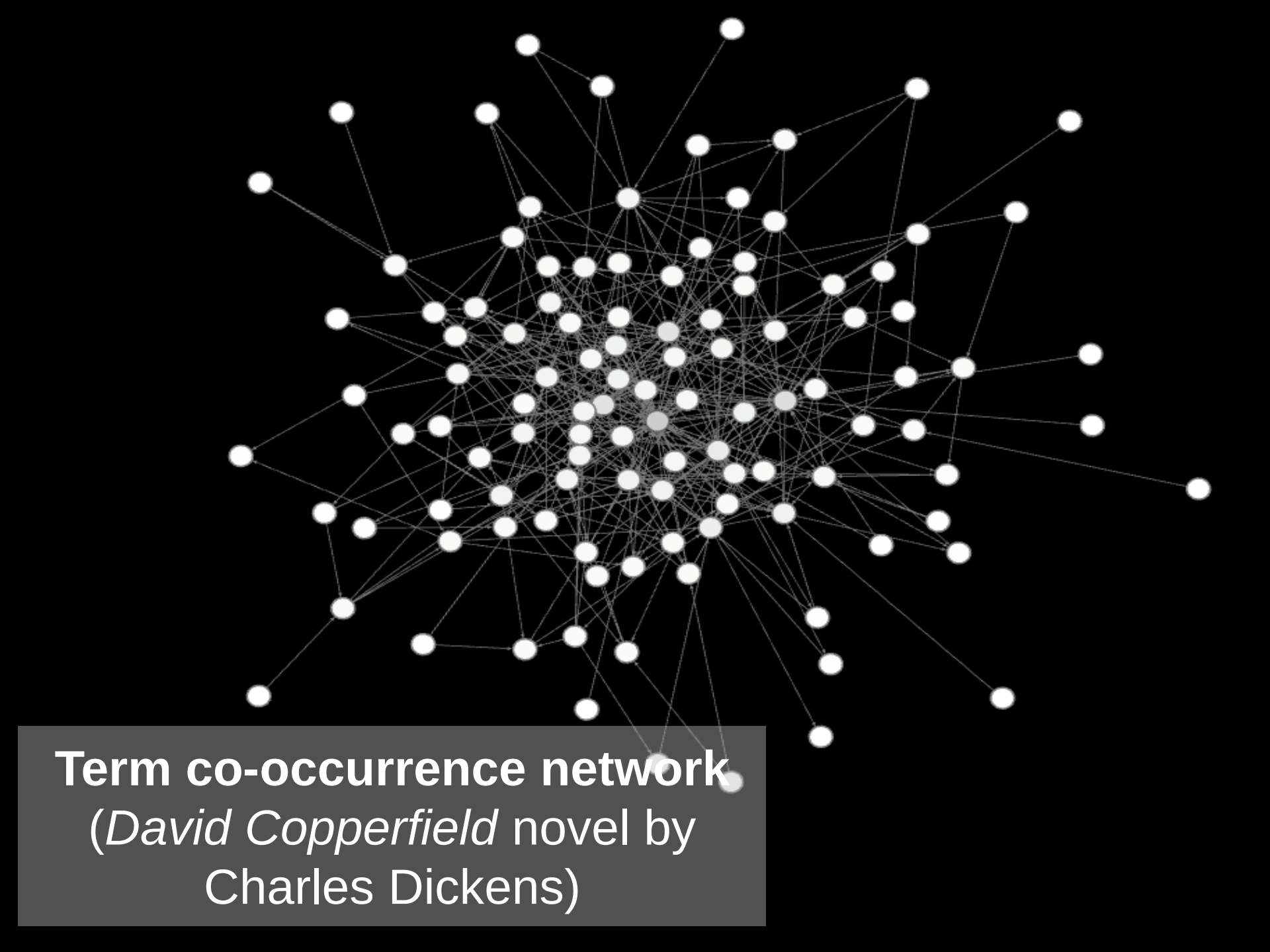
Internet graph



Source: Wikipedia

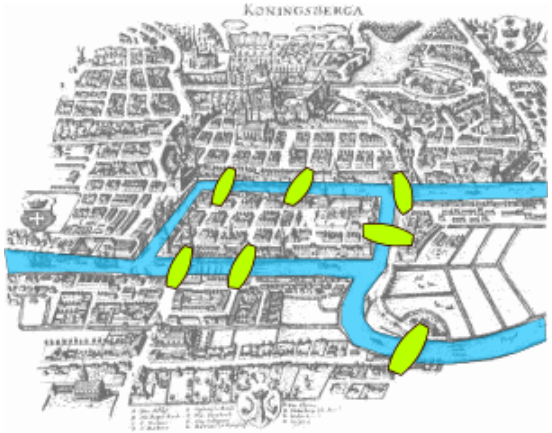


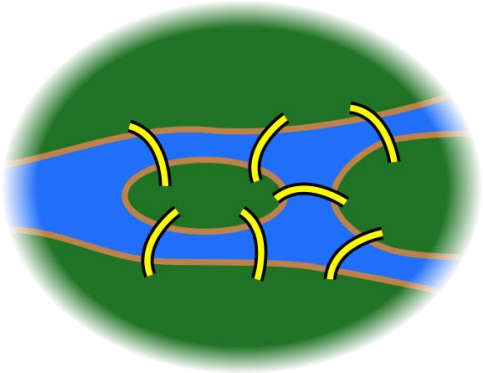
Weblogs network
(Political blogs)

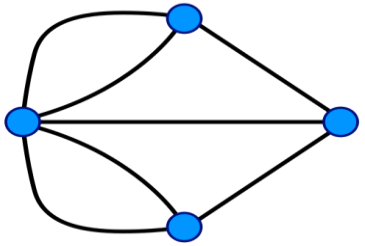


Term co-occurrence network
(*David Copperfield* novel by Charles Dickens)

Infrastructure Networks







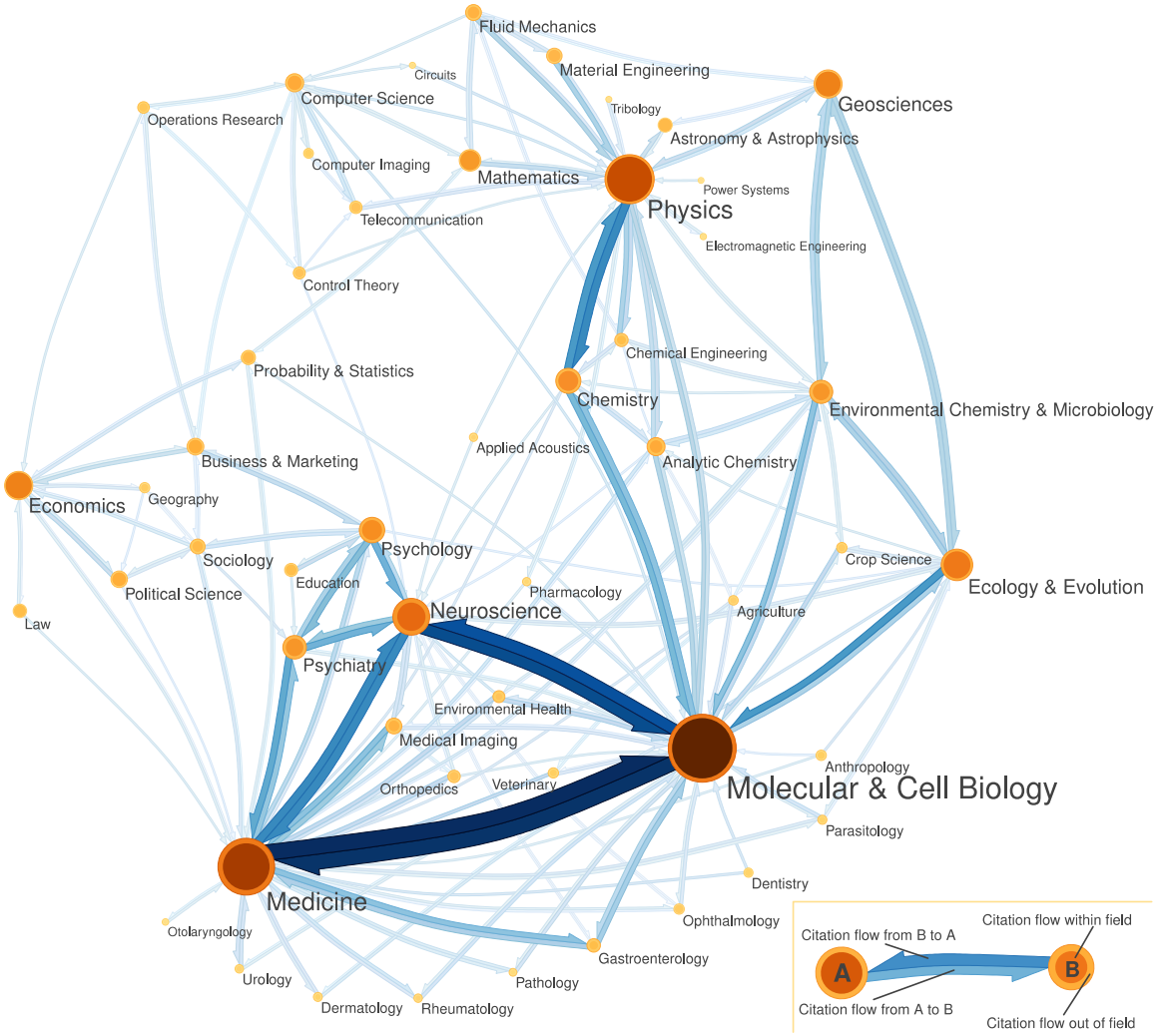


Seven Bridges of Königsberg (Kaliningrad) **[Euler, 1735]**

Devise a walk through the city that would cross each of those bridges once and only once

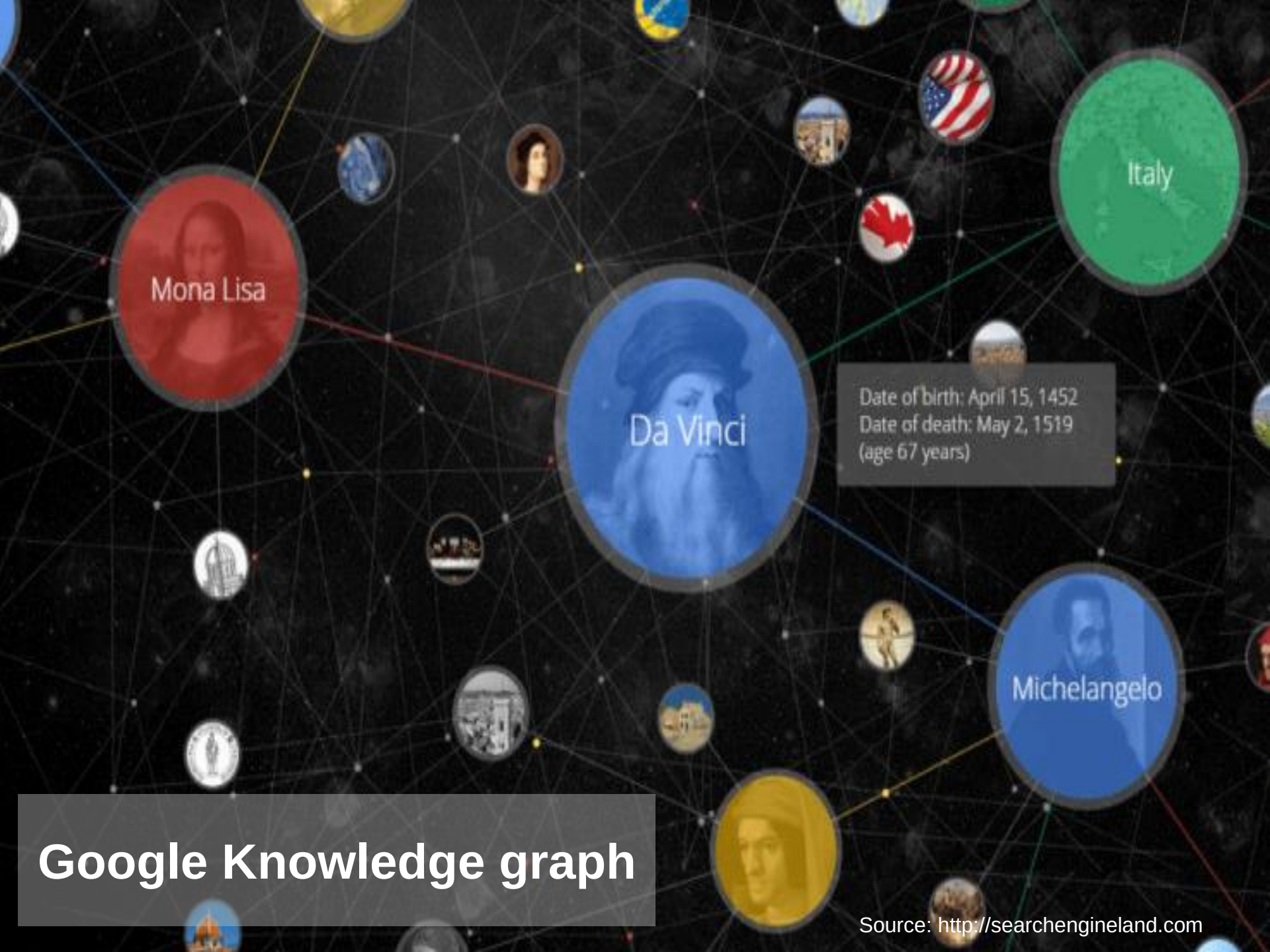
Source: https://en.wikipedia.org/wiki/Seven_Bridges_of_Konigsberg

Information Networks



Citation networks and Map of science

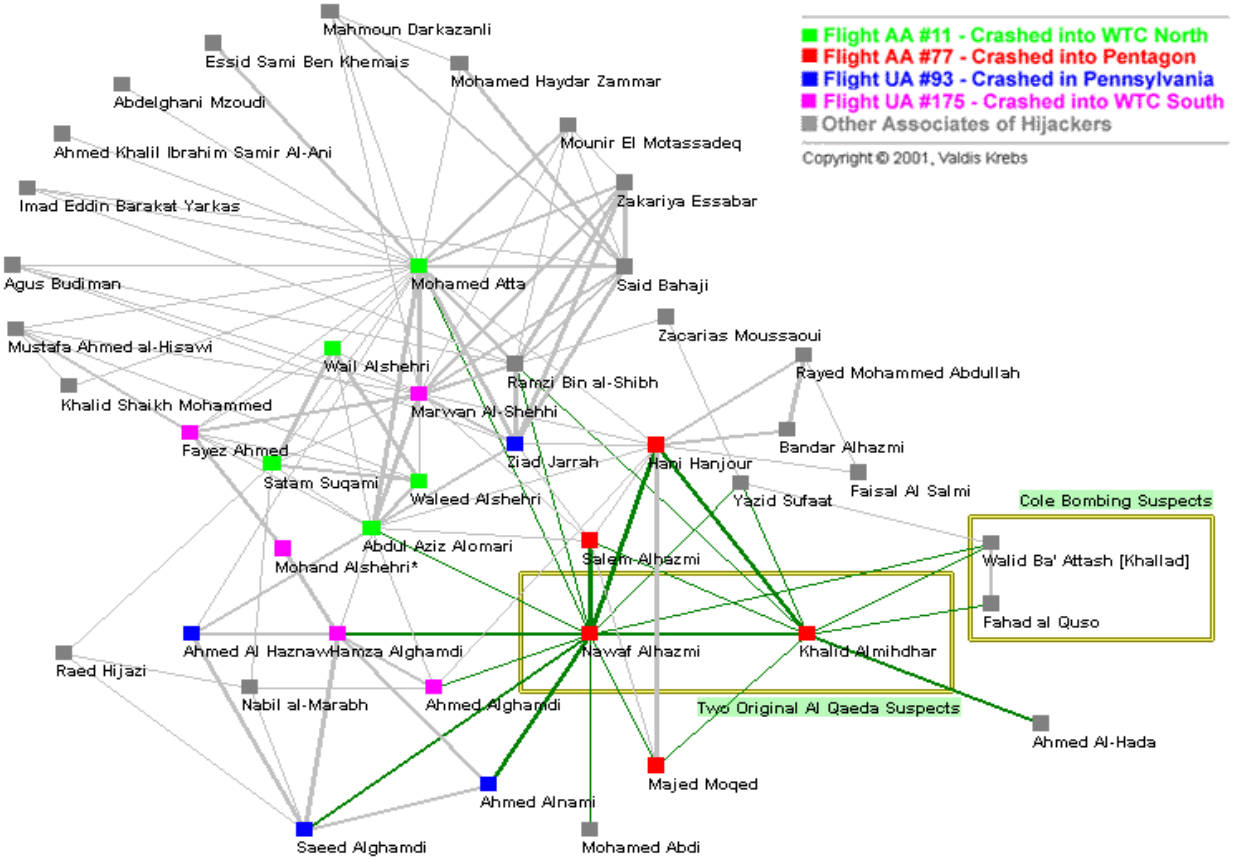
[Rosvall and Bergstrom, 2008]



Google Knowledge graph

Source: http://searchengineland.com

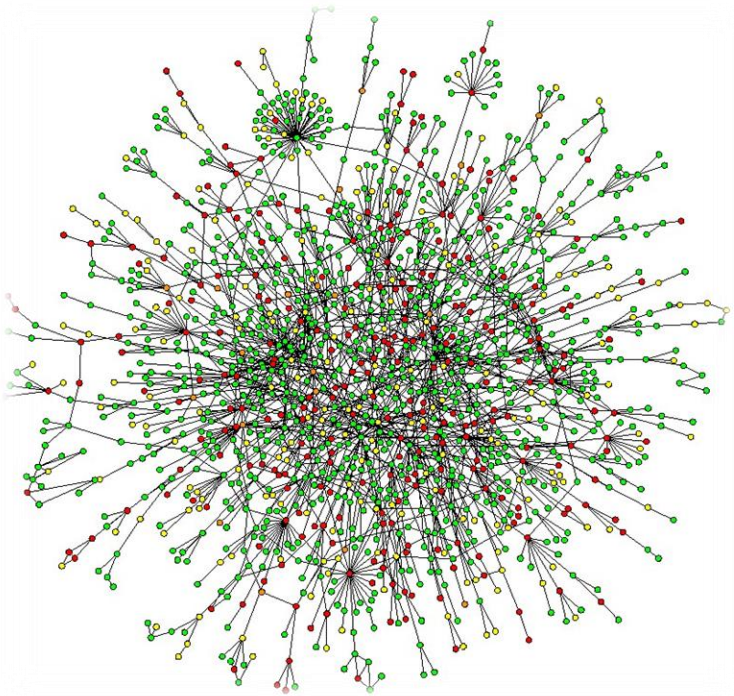
Networks of Organizations



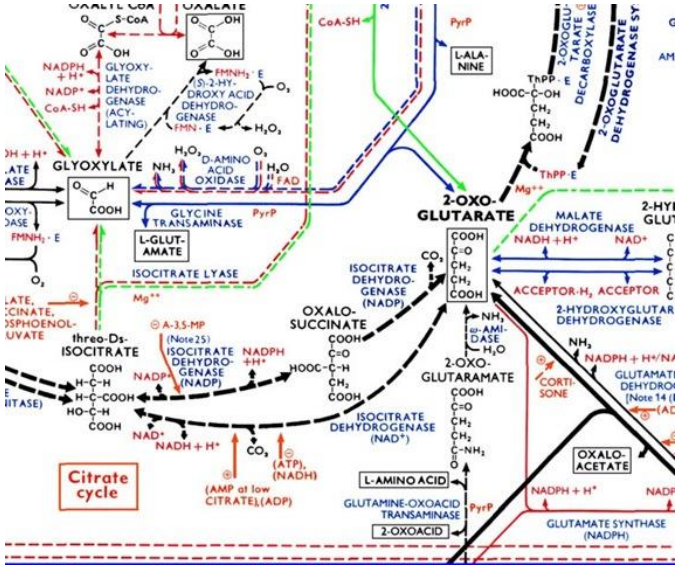
9/11 terrorist network

Source: http://www.orgnet.com/prevent.html

Biological Networks

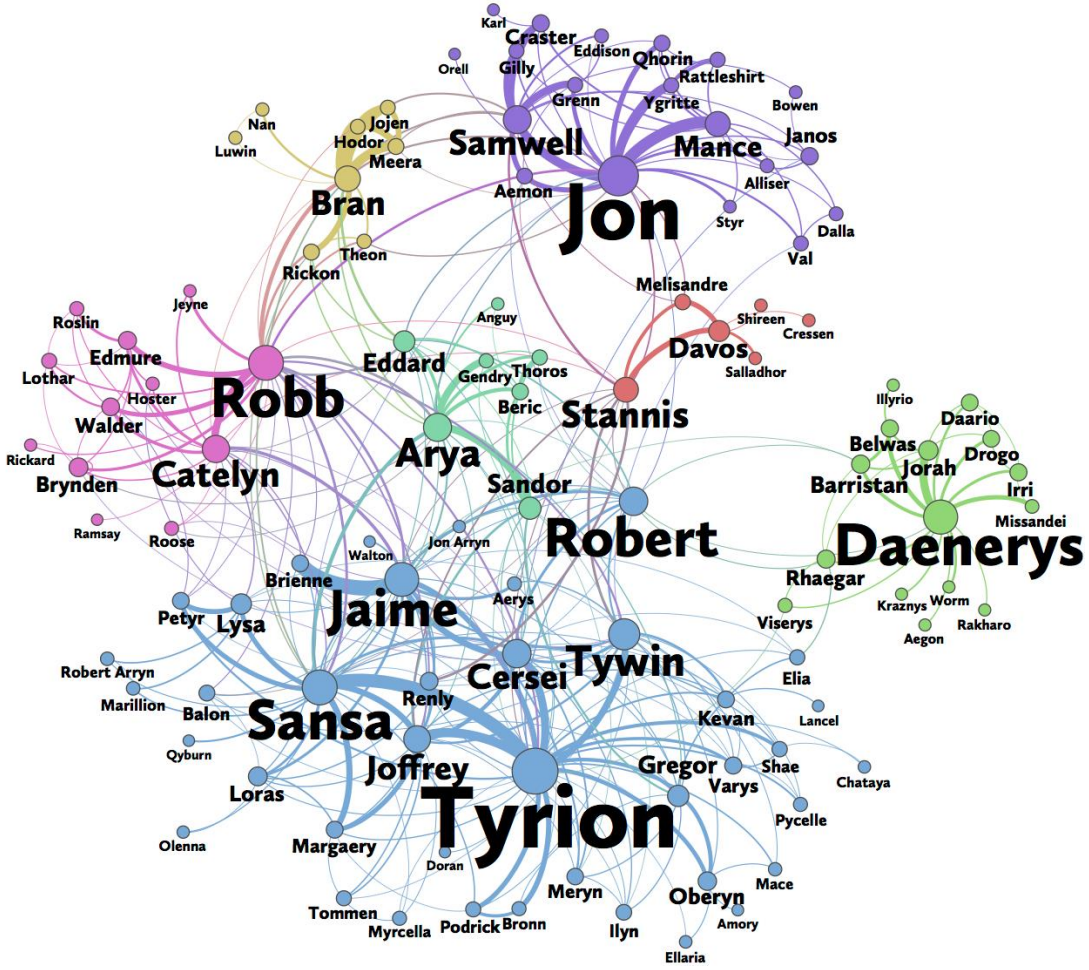


Protein-Protein Interaction Networks:
Nodes: Proteins
Edges: ‘physical’ interactions



Metabolic networks:
Nodes: Metabolites and enzymes
Edges: Chemical reactions

What Else?



In fact, anything that captures relationships between entities can be modeled as graph (any 3-way join in the DB community)

Why should I care about networks?

Why Graphs? Why Now?

- **Universal language for describing complex data**
  - Networks from science, nature, and technology are more similar than one would expect
- **Shared vocabulary (representation) between fields**
  - Computer Science, Engineering, Social Sciences, Physics, Economics, Statistics, Biology, …
  - Cross-disciplinary topic
- **Availability of big and rich data**
  - Web/mobile, bio, health, and medical
  - Computational challenges
- **Impact**
  - Social networking and social media, recommender systems, drug design, neuroscience, epidemiology, …

Networks: Scale Matters

- **Network data: Orders of magnitude**
  - 436-node network of email exchange at a corporate research lab **[Adamic, Adar, SocNets '03]**
  - 43,553-node network of email exchange at an university **[Kossinets, Watts, Science '06]**
  - 4.4-million-node network of declared friendships on a blogging community **[Liben, Nowell et al., PNAS '05]**
  - 240-million-node network of communication on Microsoft Messenger **[Leskovec, Horvitz, WWW '08]**
  - 800-million-node Facebook network **[Backstrom et al. '11]**

Web – The Lab of Humanity



The Web is a "laboratory" for understanding the pulse of humanity

Tim Berners-Lee

MIT News

ON CAMPUS AND AROUND THE WORLD

Browse or Search



FULL SCREEN

Tim Berners-Lee was honored with the Turing Award for his work inventing the World Wide Web, the first web browser, and "the fundamental protocols and algorithms [that allowed] the web to scale."

Photo: Henry Thomas

Tim Berners-Lee wins $1 million Turing Award

CSAIL researcher honored for inventing the web and developing the protocols that spurred its global use.

Adam Conner-Simons | CSAIL
April 4, 2017

Press Inquiries

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Networks: Economic Impact

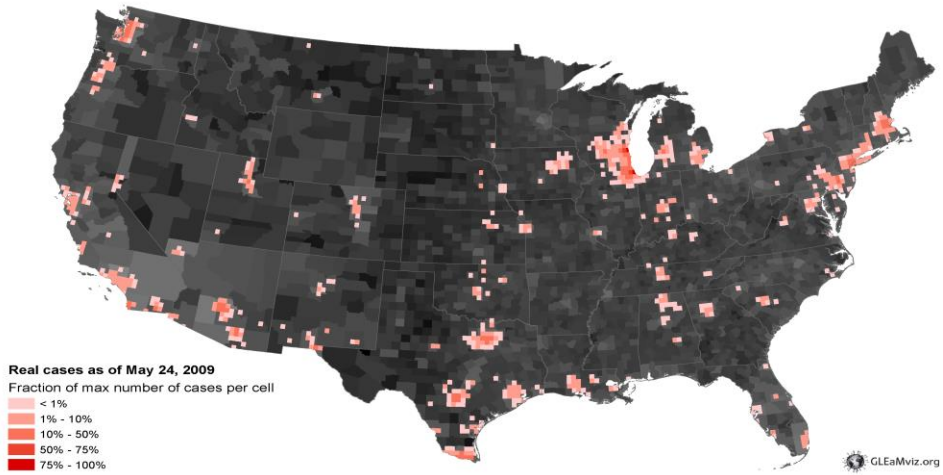


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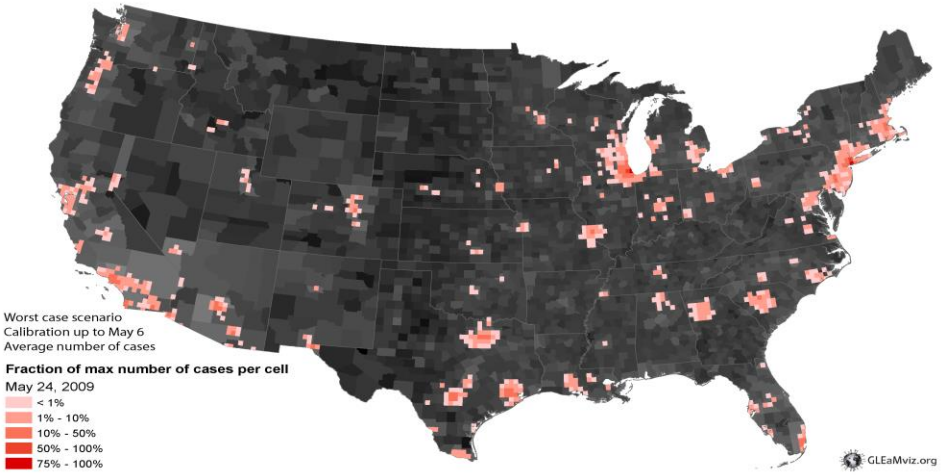
- **Google**
  Market cap: $394 billion
  (1y ago it was 300b)
- **Cisco**
  Market cap: $130 billion
  (1y ago it was 100b)
- **Facebook**
  Market cap: $201 billion
  (1y ago it was 114)

Networks: Healthcare Impact

Predicting epidemics (e.g., the 2009 H1N1 pandemic)



Real



Predicted

What Can we do With Networks?

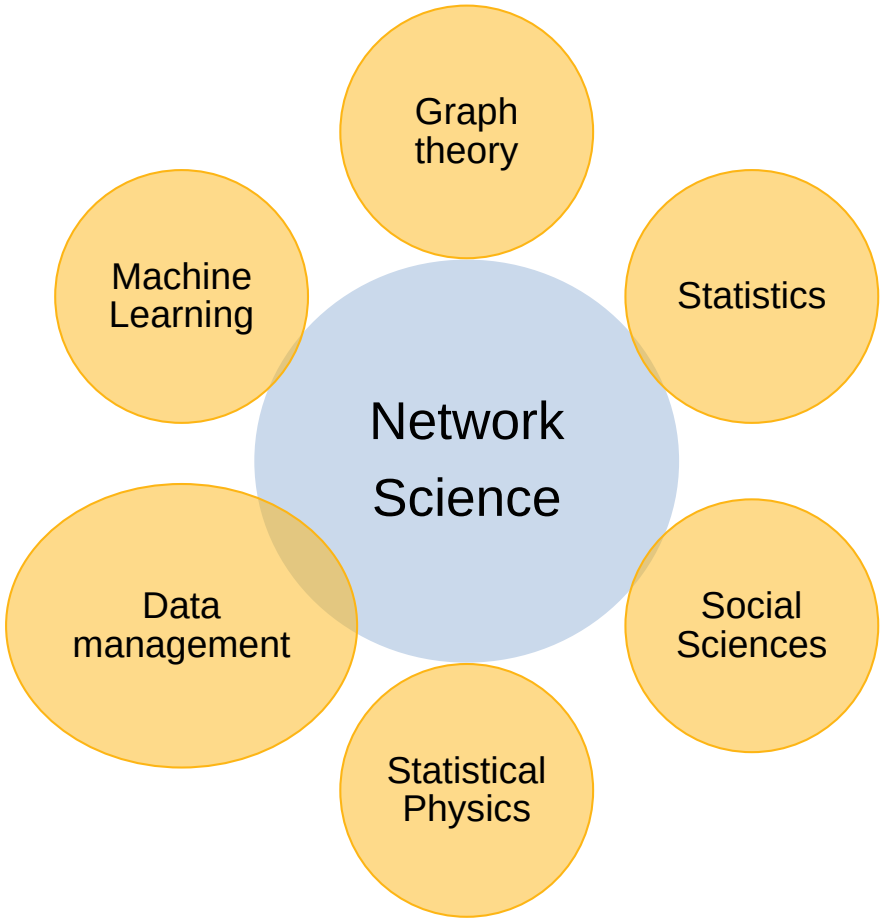
- **Communication networks**
  - Intrusion detection, fraud detection
  - Churn prediction (e.g., telecommunication providers)

- **Social networks**
  - Link prediction, friend recommendation
    - E.g., Facebook, LinkedIn
  - Social circle detection, community detection
  - Social recommendations
  - Identifying influential nodes, information spreading, influence

- **Information networks**
  - Navigational aids

Network Science Analytics

Discovering, analyzing and making sense of graph data



About this course

Reasoning about Networks (1/2)

- **What do we hope to achieve from studying networks?**
  - Patterns and statistical **properties** of network data
  - **Design principles** and **models**
  - **Understand** why real networks are organized the way they are
    - Predict behavior of networked systems
  - Utilize the **extracted knowledge** in practical applications

Reasoning about Networks (2/2)

- **How do we reason about networks?**
  - **Empirical analysis:** Study network data to find organizational principles
    - How do we **measure** and **quantify** networks?
  - **Mathematical models:** Graph theory and statistical models
    - Models allow us to understand behaviors and distinguish **surprising** from **expected** phenomena
  - **Algorithms** for analyzing graphs
    - Hard computational challenges (scale and complexity of the underlying networks)
    - Unsupervised vs. supervised algorithms

Learning Objectives

- Introduce students to the field of **graph mining** and **network analysis** and **machine learning on graph**
  - Cover a wide range of topics, methodologies and related applications
  - Hands-on experience on dealing with graph data and graph mining tasks

- By the end of the course, we expect
  - To have a thorough understanding of various graph mining and learning tasks
  - Be able to analyze large-scale graph data
  - Formulate and solve problems that involve graph structures

Course Logistics

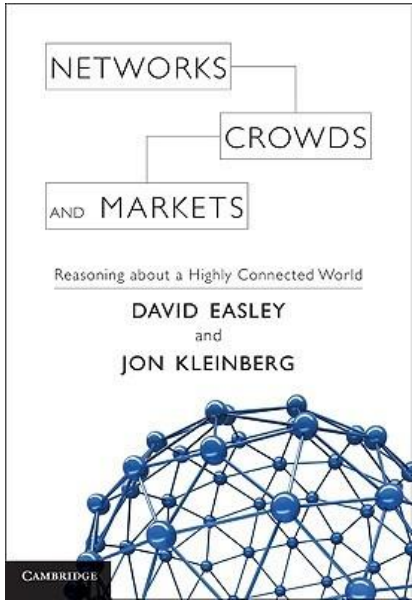
- **Website**
  - http://fragkiskos.me/teaching/NGSA-W20/
  - Information about the course, schedule, reading material
  - Resources (helpful for the project)

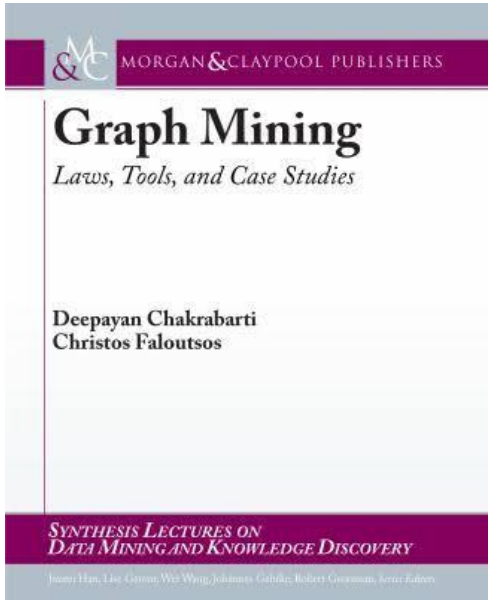
- **Piazza for Q&A** (you need to register)
  - https://piazza.com/centralesupelec/winter2020/ngsa
  - Please, participate and help each other
  - All announcements will be posted there
  - Also, lecture slides and assignments
  - Use key to enroll: **ngsa-20**

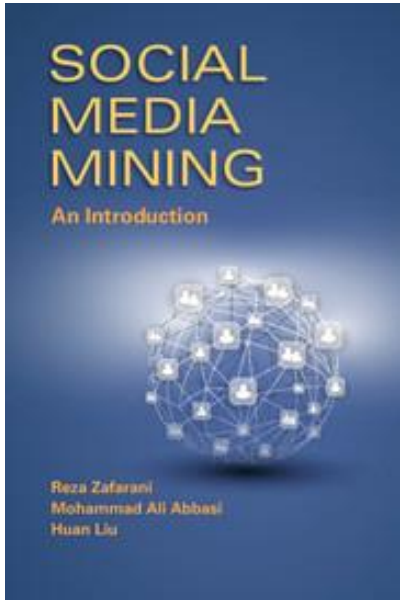
Prerequisites

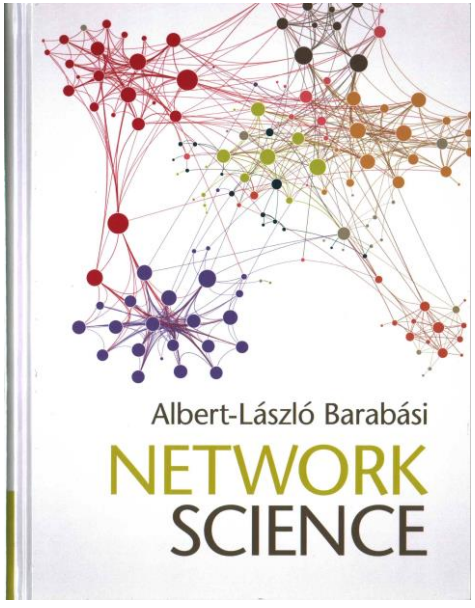
- Basic knowledge of
  - **Graph theory** and **algorithms**
  - **Linear algebra**
  - **Probabilities and statistics**
- Be familiar with basic tasks in **machine learning** / **data mining**
  - E.g., Clustering, classification
- Programming is necessary
  - Python (or any other language of your preference)

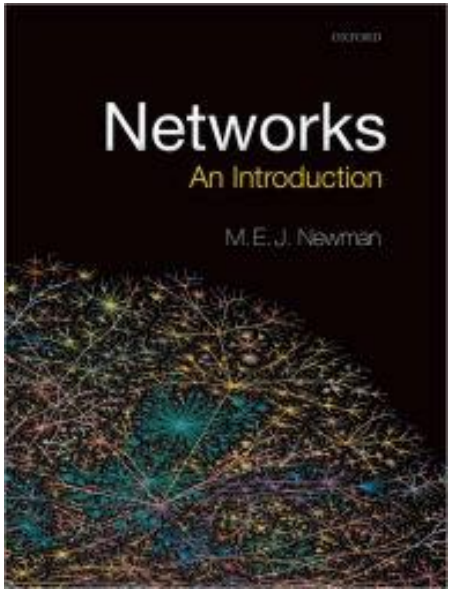
Reading Material











- The books are publicly available in electronic form (except the “Networks: An Introduction)
  - Pointers to chapters for every lecture (see the website)
- Research articles for some topics

Structure of the Course

Three components:

1. Lectures
   - First half of each session (except the first two lectures)
2. Lab sessions
   - Hands-on experience on graph mining algorithms
     - Some of the algorithms will be implemented from scratch
   - Need to install software and to experiment in class
   - Labs will not be graded, but will help you to further understand the material presented in the lectures
   - 4-5 lab sessions in total
3. Data challenge and project

Coursework and Grading

	Weight	Details
Data challenge (teams of 3-4 students)	40%	• Deal with a real network science task • Kaggle competition • Deliver short report and code
Project (teams of 3-4 students)	60%	• Proposal • Final report • Presentation (oral in the class or poster session)

- Small adjustments might be done in the weights of the coursework
- A detailed description of the project will be provided soon

Course Project (1/2)

- Main component of the evaluation of the course
  - Experimental evaluation of algorithms on an interesting graph dataset
  - Empirical comparison of algorithms for a specific task
  - A theoretical project that considers an algorithm / model and derives rigorous results about it
  - A new algorithm for a graph mining task and evaluation
  - Formulation of an interesting problem using graph mining techniques, algorithm and evaluation
  - New network dataset, exploration and an application (make the dataset available)
  - Efficient implementation of an algorithm and experimental evaluation
  - Your ideas along the lines of your own research
- Ideally, combination of experimentation on real/artificial data and theoretical analysis
- The final report should have the structure of a research article

Course Project (2/2)

- Please see the project section on piazza [will be added soon]
- Check the **Resources** section on the website for ideas about interesting datasets, tasks, data challenges
  - Start thinking about the project as early as possible
  - We will help with ideas and mentoring
- Presentation in class (or poster session) at the end of the course

It will be fan ☺

Software Tools

- We strongly advise to use **Python**
  - **NetworkX** library
  - **igraph** library (also for C++ and R)
- **Snap** library
  - C++ and Python
- **Gephi**, **Jung** and **graph-tool** for network visualization
- See the **Resources** section

Some Personal Notes ☺

- Come to the class (I know, it's very early in the morning ☺)
  - Please ask questions, participate in discussions on piazza
- Check out the additional suggested material on the website
  - Special topics, **research-oriented** course
  - Search the web, google is your friend!
  - For every topic covered in the class, you can find the original research articles – take a look on them
  - Typically, the suggested reading material is overlapping – read selectively
- Play with software tools
  - This is the actual goal of the assignments
- Give us your feedback!

Topics that will be covered

Schedule (Subject to Change)

http://fragkiskos.me/teaching/NGSA-W20/

Schedule and Lectures

The topics of the lectures are subject to change (the following schedule outlines the topics that will be covered in the course). The slides for each lecture will be posted in piazza just before the start of the class. **The due dates of the assignments/project are subject to change.**

Week	Date	Topic	Material	Assignments/Project
1	February 7	○ Introduction to network science and graph mining ○ Graph theory and linear algebra recap; basic network properties	Lecture 1A Lecture 1B	
2	February 14	○ Random graphs and the small-world phenomenon ○ Power-law degree distribution and the Preferential Attachment model	Lecture 2A Lecture 2B	
3	February 21	○ Time-evolving graphs and network models ○ Centrality criteria and link analysis algorithms	Lecture 3A Lecture 3B	
4	February 28	○ Graph clustering and community detection	Lecture 4	**Project proposal due on March 1**
5	March 6	○ Node similarity and link prediction ○ Graph similarity	Lecture 5A Lecture 5B	**Data challenge out**
6	March 13	○ Representation learning in graphs ○ Graph sampling and summarization	Lecture 6A Lecture 6B	
7	March 20	○ Epidemic processes and cascading behavior in networks ○ Influence maximization in social networks	Lecture 7A Lecture 7B	**Data challenge due on March 22**
8	April 10	Project presentations		**Project final report due on April 10** **Project poster session or presentations**

Networks or Graphs?

- **Network** often refers to real systems
  - Web, Social network, Internet Metabolic network

  Language: network, node, link

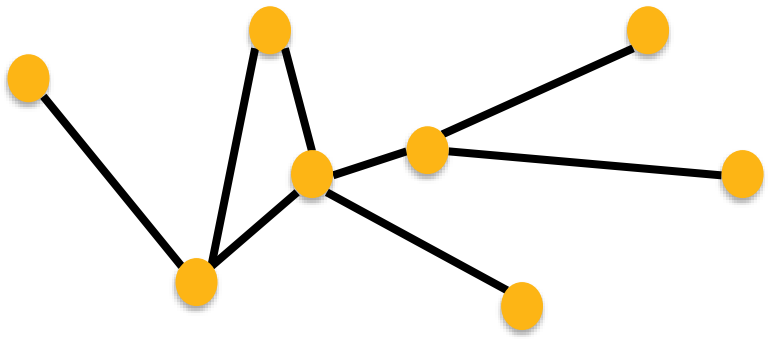
- **Graph** is mathematical representation of a network (a model)
  - Web graph, Social graph (a Facebook term)

  Language: graph, vertex, edge

We will try to make this distinction whenever it is appropriate, but in most cases we will use the two terms interchangeably

Basics in Graph Theory and Linear Algebra

In today's lecture (Part B)



$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Adjacency matrix

- **Objects:** nodes, vertices ***V***
- **Interactions:** links, edges ***E***
- **System:** network, graph ***G = (V, E)***

Graph-theoretic algorithms

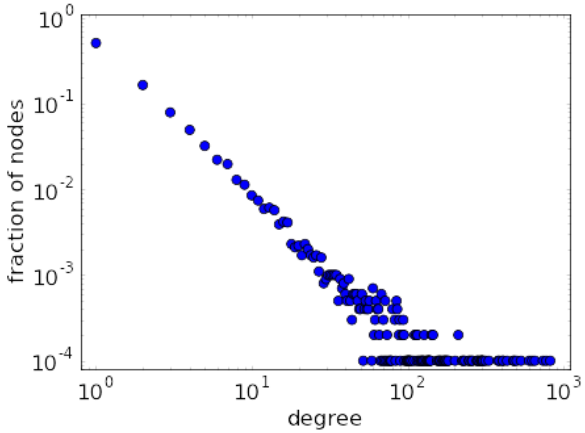
- E.g., graph concepts, types of graphs, subgraphs, traversal, shortest paths, connectivity, complexity issues

Linear algebra

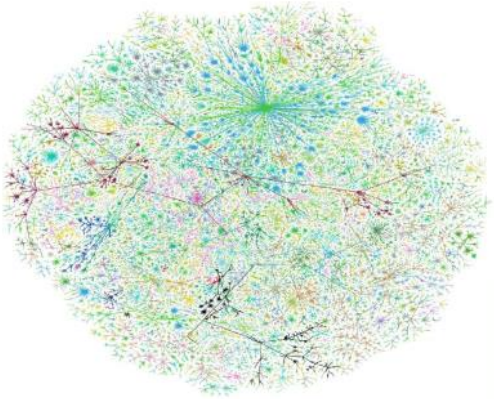
- E.g., matrix-based graph representations, matrix decomposition, properties of the adjacency matrix, Laplacian matrix, spectral graph theory, graph spectrum

Patterns and Graph Generative Models

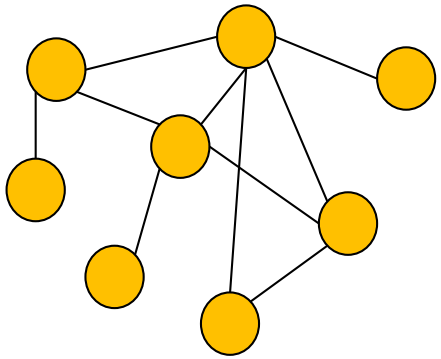
Q1: How does a real-network look like?
Q2: Properties, patterns, deviation from **randomness**?



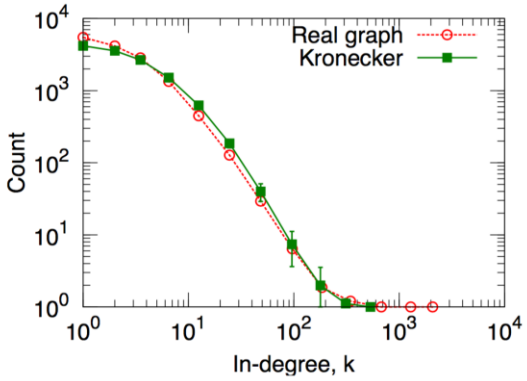
Q3: Can we generate artificial networks that are similar to real ones?



Given a **real** graph
(e.g., Facebook social graph)

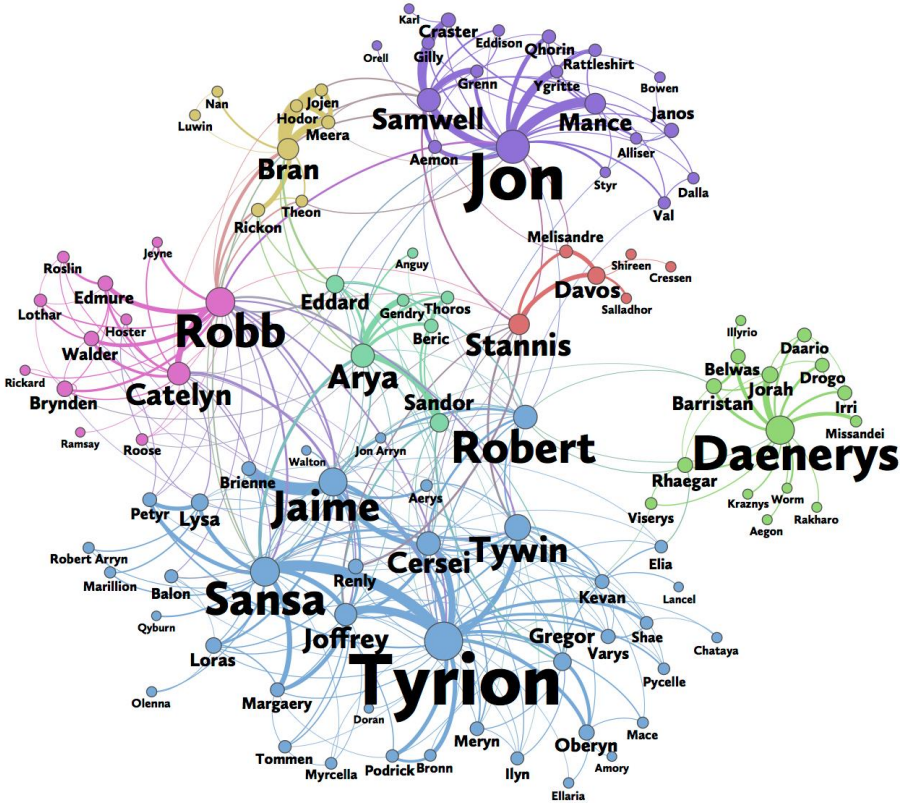


Generate an **artificial** graph
(e.g., model the formation process)



Fit some properties
(e.g., same degree distribution)

Centrality and Ranking in Networks

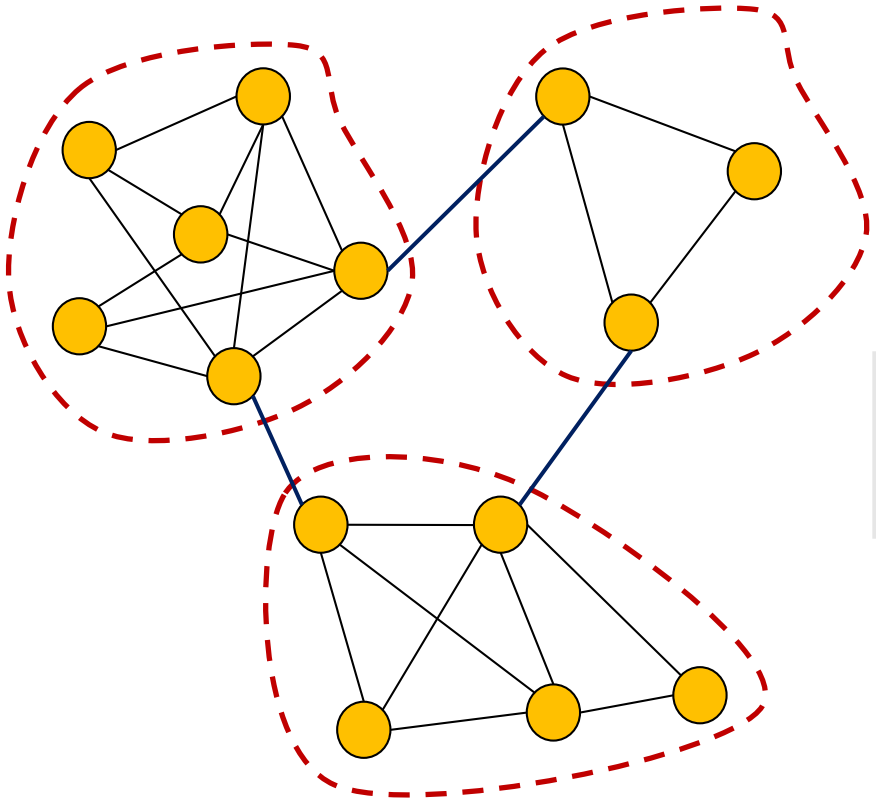


Q: How to determine the importance of a node in the graph?

- **Centrality** criteria (e.g., degree, closeness, betweenness)
- HITS and PageRank algorithms
- Scalability issues

Source: https://www.macalester.edu/~abeverid/thrones.html

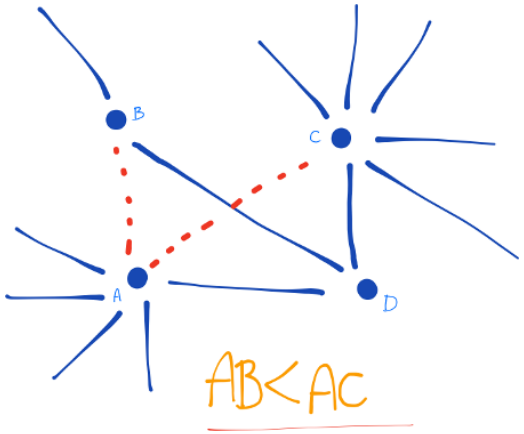
Graph Clustering - Community Detection



Example graph with three communities

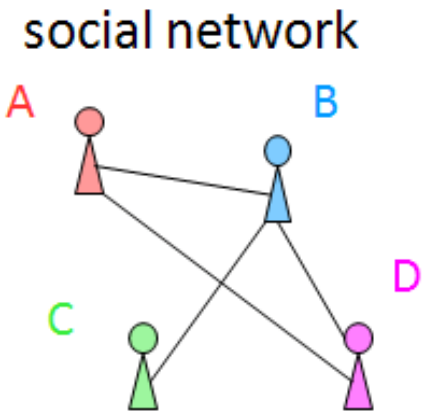
- **[Models and definitions]** How a community in graphs looks like?
- **[Algorithms]** How can we extract the inherent communities?
- **[Patterns in large networks]** What is happening in large-scale networks?

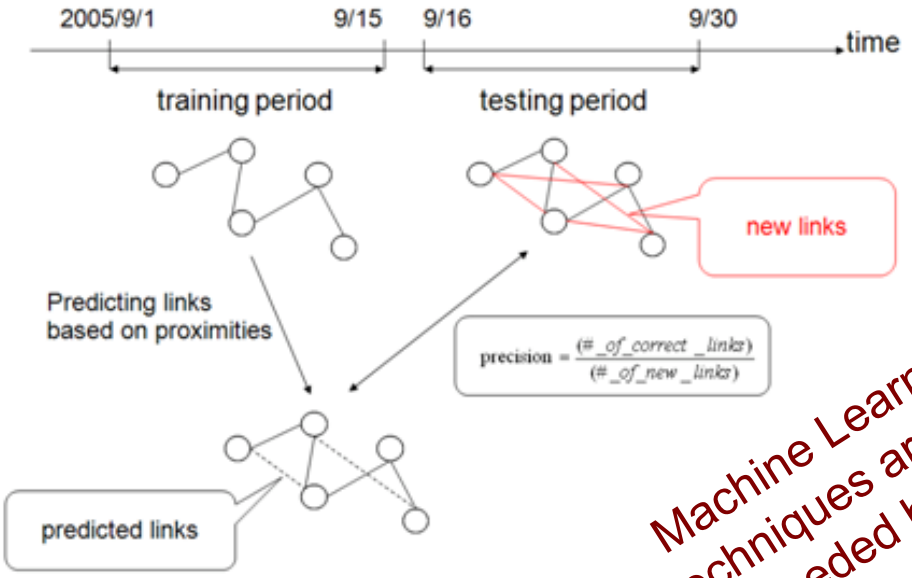
Node Similarity and Link Prediction



- How similar are two nodes in the graph?
- Can we utilize this similarity to predict missing edges (links)?

Applications?



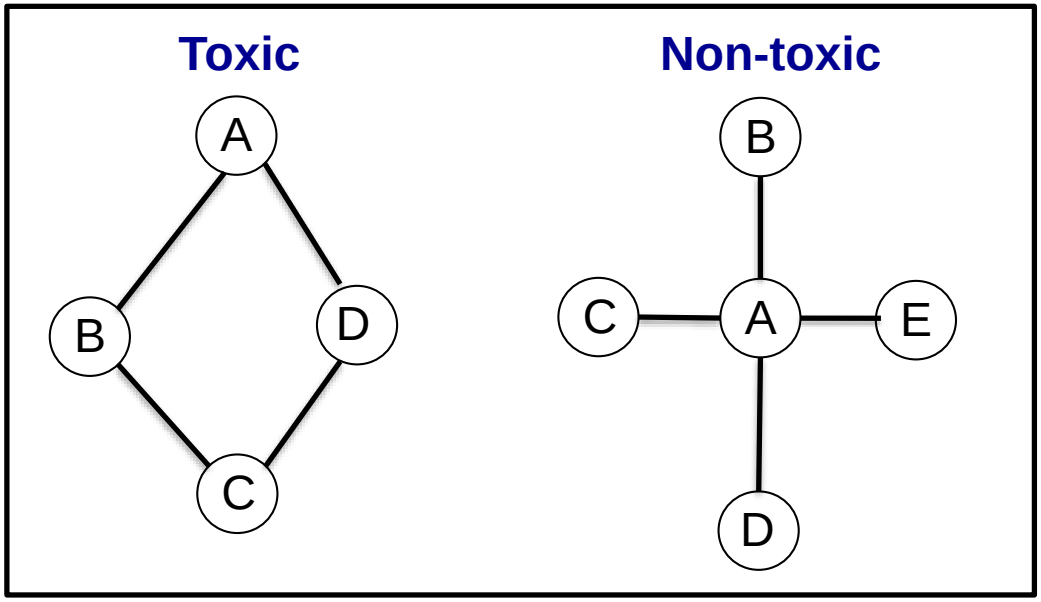


Machine Learning techniques are also needed here!

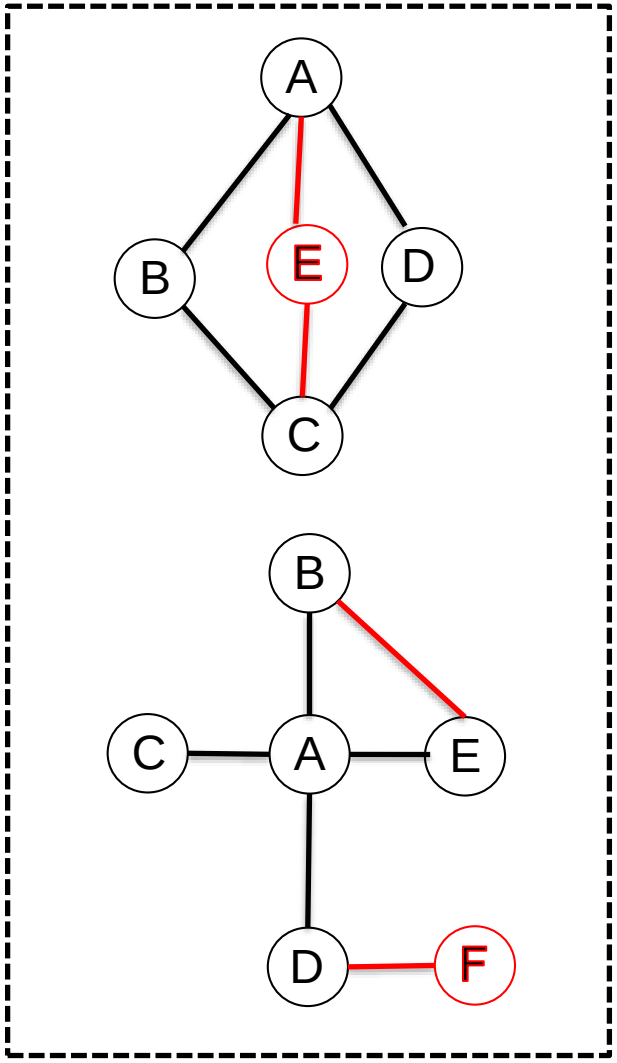
Source: http://be.amazd.com/link-prediction/
http://www.net.c.titech.ac.jp/research.html

Graph Similarity and Classification

Dataset of **known** molecules

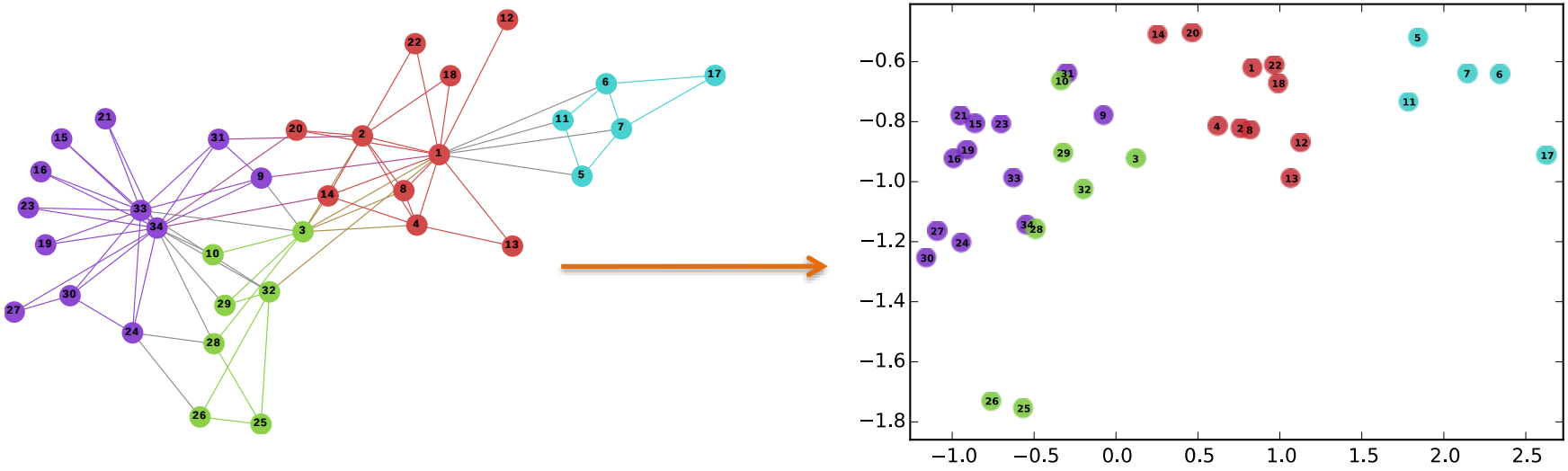


Unknown molecules



Task:
Given a set of molecules that are either toxic or non-toxic
Predict the class of unknown molecules

Representation Learning on Graphs



Input graph

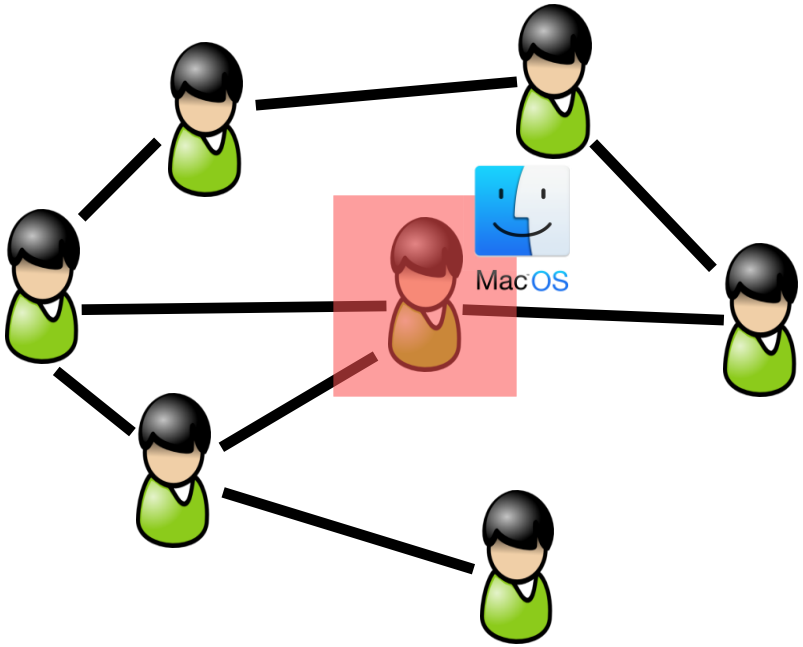
Learn latent representation

Applications: clustering, classification, link prediction, …

Also known as node embedding techniques (e.g., DeepWalk, node2vec)

[Perozzi et al., KDD '14]

Influential Nodes and Influence Maximization



[Viral marketing] How to organize an effective product promotion campaign?

[Opinion dynamics] How do opinions/rumors spread?

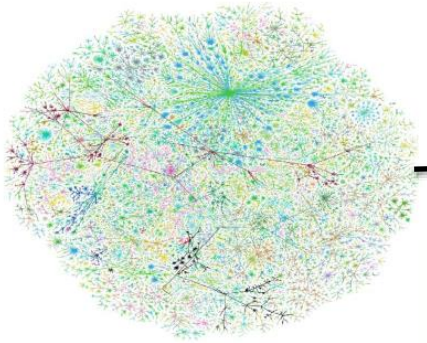
[Epidemiology] How do viruses/diseases propagate?

- Detection of influential nodes (spreaders) in networks
- Influence maximization algorithms
- Epidemic processes in networks

[Prakash, Ramakrishnan, KDD '16]

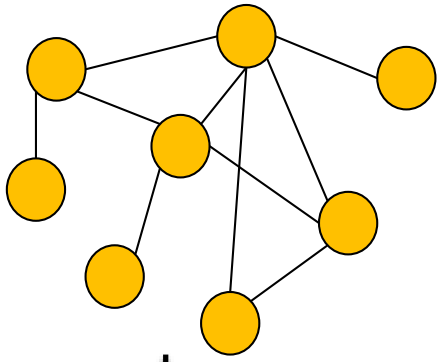
Graph Sampling and Summarization

Original graph G



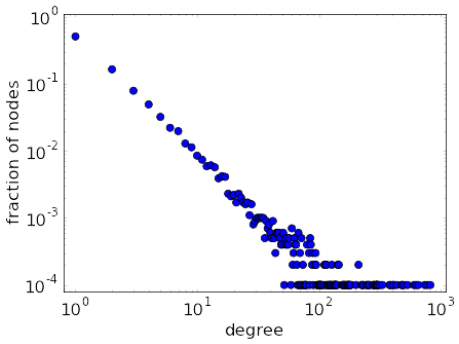
Sampling

Sample subgraph G_S
(or sample of nodes/edges)

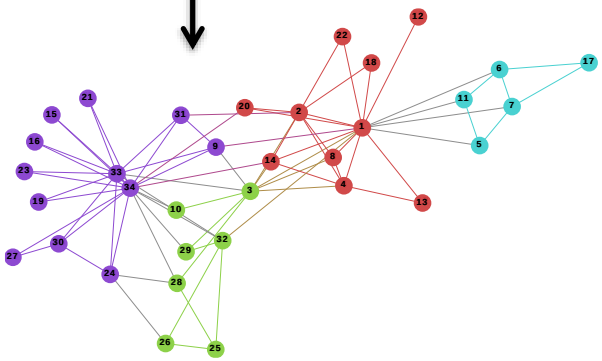


Reduce size

- Estimate graph properties quickly
- Extract smaller subgraph that preserves the properties of the original graph
- Algorithms run faster

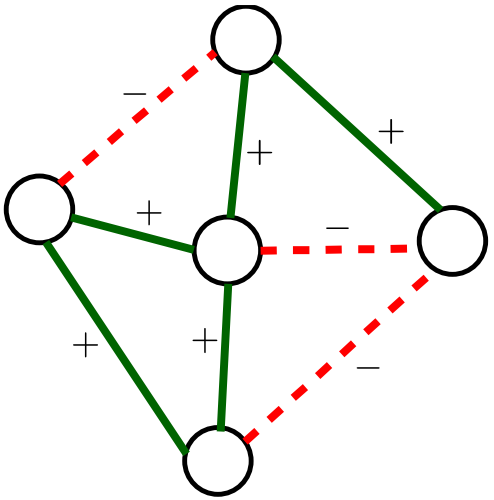


Estimate characteristics of the graph



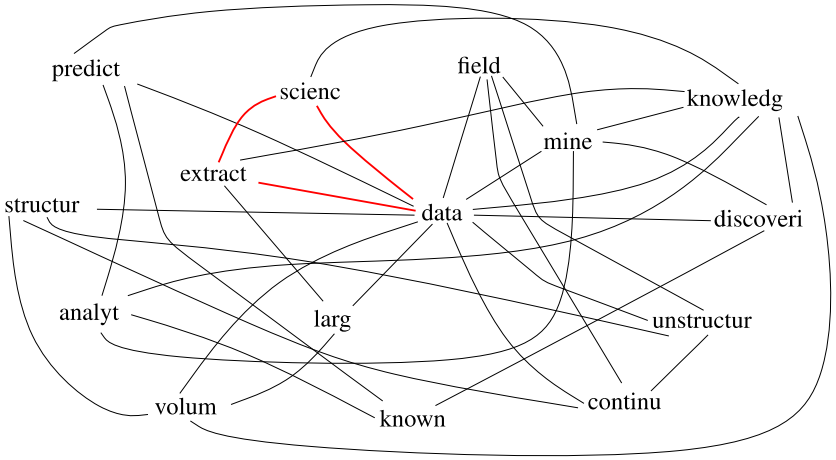
Extract the community structure

Rich Network Structures



Signed graphs

Data Science ~~is the~~ extraction of knowledge from large volumes of data that are structured or unstructured which is a continuation of the field of data mining and predictive analytics, also known as knowledge discovery and data mining.



Text-based information networks

Other topics (last lecture – if time permits**):**

- Probabilistic (uncertain) graphs (probabilities on the edges)
- Multilayer networks (various types of relationships among nodes)
- Location-based systems (e.g., Foursquare)
- Biological networks

Lecture 1 – Part B

- Basics in
  - Graph theory
  - Linear algebra and spectral graph theory

Thank You!



Source: http://www.pacificworld.com/greece/